[INCH–POUND]
A–A–59832/5
5 March 2011

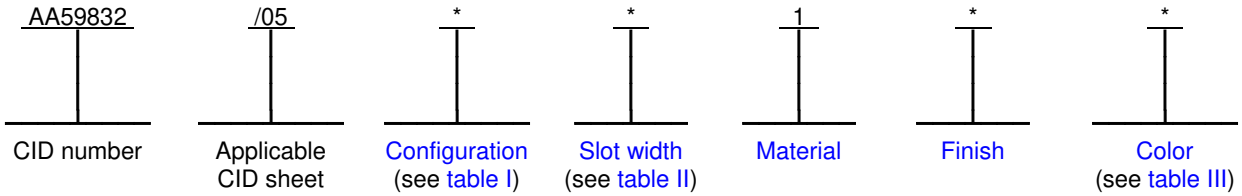
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

EXTRACTOR, ELECTRICAL CARD, METAL, NON–LOCKING, SINGLE–SIDED ACTUATION, NARROW PROFILE, ANGLED ACTUATING HANDLE FOR .032, .063, .094, .125, AND .156 INCH THICK CIRCUIT CARD ASSEMBLIES

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring extractors described herein shall consist of this document and the latest issue in effect of A–A–59832.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES herein).



Example: AA59832/05 K42LC is the PIN for a left handed angle handle aluminum extractor designed for mounting on a circuit card assembly with a printed board thickness of .125 inch (3.20 mm) in width. The corrosion protection finish applied to the extractor is clear chemical film and the handle has a sensitive electronic device (SED) symbol.

SALIENT CHARACTERISTICS.

Performance. Extractors shall be capable of initiating the removal of a circuit card assembly from its installed position. The mechanical advantage of all extractor configurations on this CID specification sheet is approximately 4.2:1.

Interface and physical dimensions. The extractors supplied to this CID specification sheet shall be as specified herein and meet the general requirements specified in CID A–A–59832.

Configuration. The configuration of an extractor shall be as specified in table I and on figures 1 through 6.

TABLE I. Configurations.

Configuration	Applicable figures	Extractor description
K	1 and 2	Left hand with SED symbol on the angled handle
L	1	Left hand with angled handle
M	3	Left hand with bent leg and angled handle
N	3 and 4	Left hand with bent leg and SED symbol on the angled handle
P	4 and 5	Right hand with bent leg and SED symbol on the angled handle
Q	5	Right hand with bent leg and angled handle
R	6	Right hand with angled handle
S	6 and 2	Right hand with SED symbol on the angled handle

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Slot width. The available slot widths needed to accommodate various printed board thicknesses for the extractors covered by this CID specification sheet are specified in table II. The slot width designator shall be included in the PIN.

TABLE II. Slot width configurations and dimensions. 1/

Slot width designator	Slot width dimension "A" 2/ Inch (mm)	Accommodates printed board thickness Inch (mm)	Roll pin width Inch (mm)	Handle width dimension "H" Inch (mm)
1	.040 (1.02)	.032 (0.8)	.187 (4.75)	.25 (6.4)
2	.075 (2.03)	.063 (1.6)	.187 (4.75)	
3	.105 (2.79)	.094 (2.4)	.250 (6.35)	
4	.142 (3.56)	.125 (3.2)	.250 (6.35)	.29 (7.4)
5	.165 (4.19)	.156 (4.0)	.312 (7.94)	

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for general information only.
2/ Tolerance is +.010, –.000 inch (+0.25, –0.000 mm).

Material. The material used in the extractor body shall be as specified in A–A–59832. The applicable material type designators for this CID specification sheet are "1" (aluminum alloy 6061, temper T6), "2" (aluminum alloy 5052, temper H32), or "7" (Stainless steel, type 301 or 304). The material designator shall be included in the PIN.

Material thickness. The aluminum or stainless steel thickness shall be .063 ±.002 inch (0.23 ±0.05 mm).

Roll pin. A stainless steel roll pin is furnished with each extractor. The roll pin is .09375 inch (2.38 mm) diameter by the lengths specified in table II.

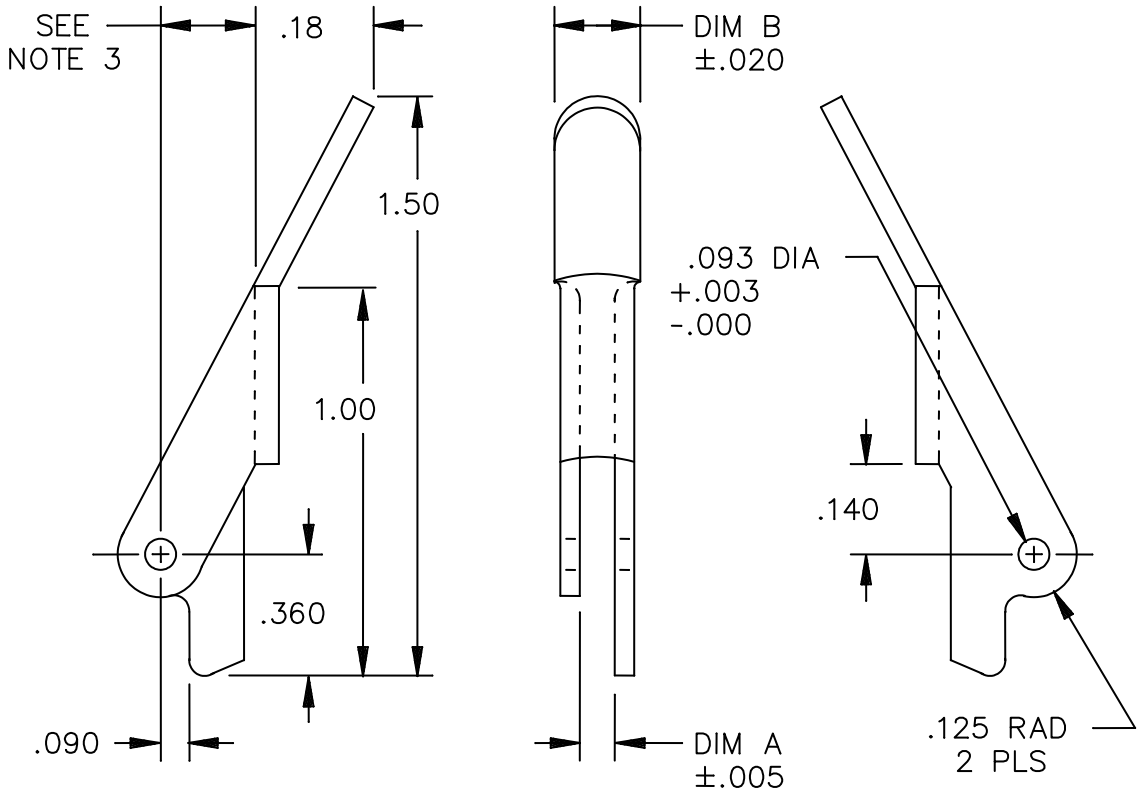
Finish. The finish designator shall be as specified in A–A–59832. The finishes available for this CID specification sheet are as follows: "A" (anodize) or "L" (low resistance chemical film) for extractors fabricated of aluminum alloys and "P" (passivation treatment) extractors fabricated of stainless steel.

Color. The color designator shall be as specified in A–A–59832. The colors available for this CID specification sheet shall be "C" (clear) or "G" (gold) for low resistance chemical film and "C" (clear) for passivation treatment. The color of anodized extractors shall be specified in table III. The color designator shall be included in the PIN.

TABLE III. Colors of anodized extractors.

Designator	Color	Designator	Color
B	Black	R	Red
C	Clear	U	Blue
D	Green	V	Violet (purple)
E	Gray	W	Brown
F	Orange	Y	Yellow

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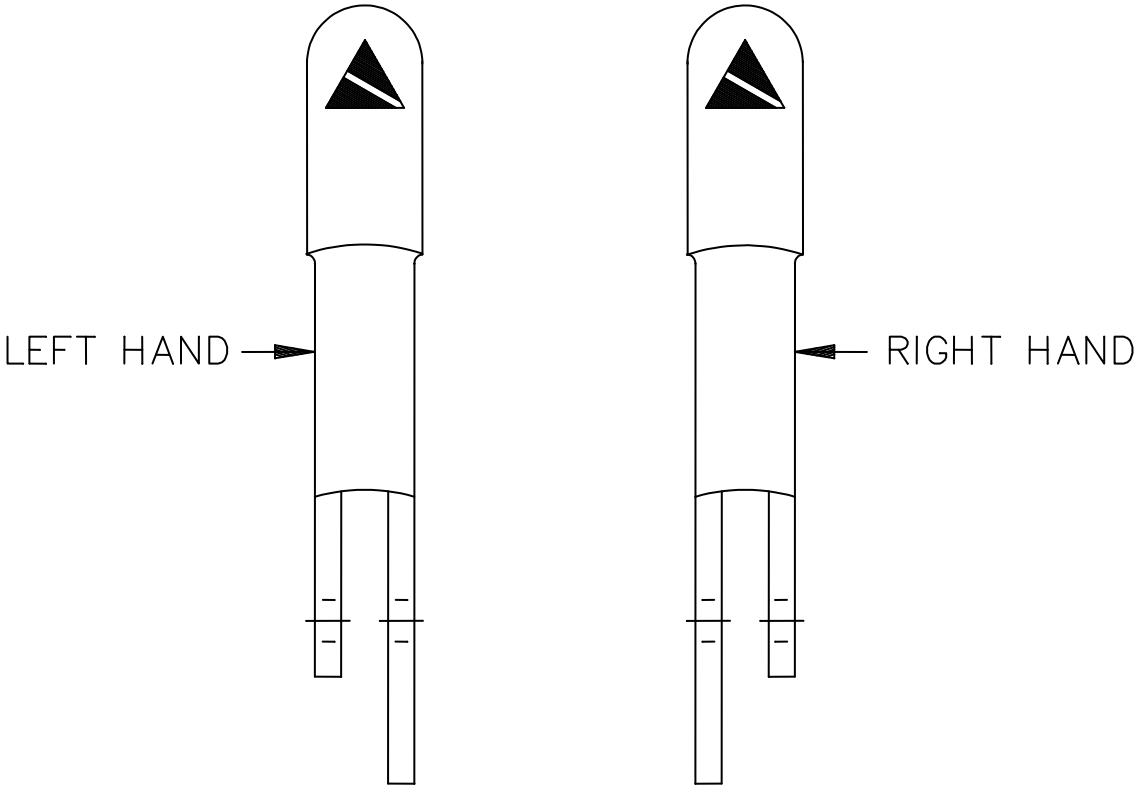
BOARD EDGE VIEW

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.020	0.51	.093	2.36	.140	3.56	.360	9.14
.005	0.13	.090	2.29	.125	3.18	.18	4.57	1.00	25.4
								1.50	38.1

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Depth dimension is .29 inch (7.4 mm) for extractors from CAGE 18915 and 5GB68 and .27 inch (6.9 mm) from CAGE 61081.

FIGURE 1. Configuration L dimensions (left hand).

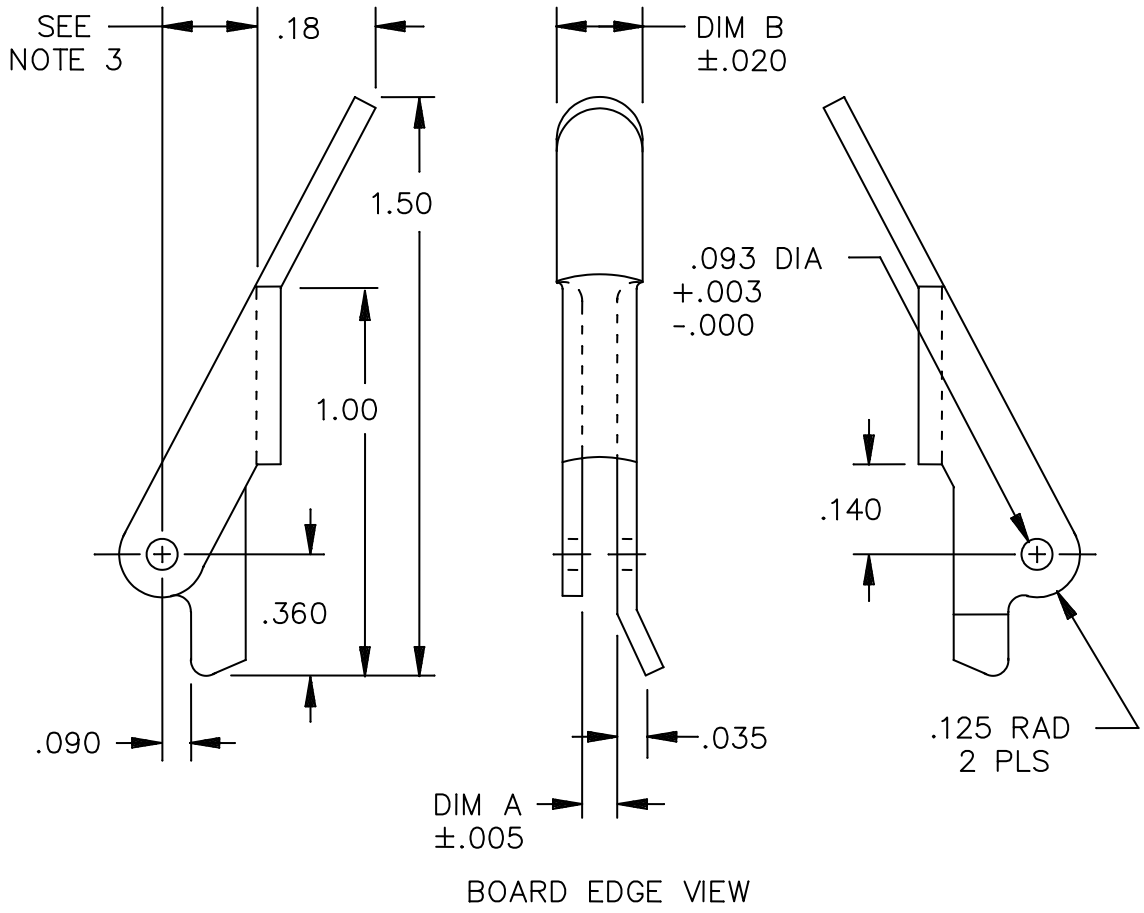
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. The SED symbol displayed on the extractor handles shall be in accordance with A–A–59832.
2. Unless otherwise specified, a monochromatic reproduction of the symbol displayed in A–A–59832 using any color that contrasts with the color of the extractor handle may be used.

FIGURE 2. Configurations K and S with SED symbol on extractor handle (board edge view).

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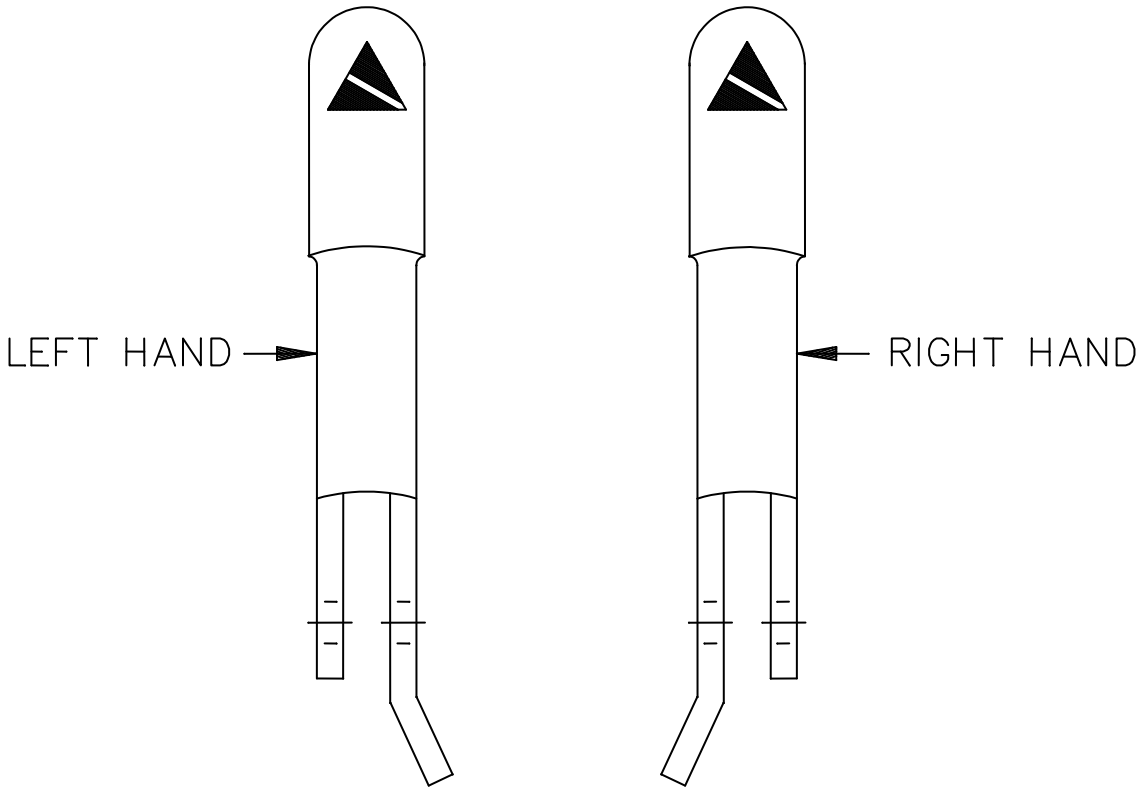
Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.020	0.51	.090	2.29	.125	3.18	.360	9.14
.005	0.13	.035	0.89	.093	2.36	.140	3.56	1.00	25.4
						.18	4.57	1.50	38.1

NOTES:

1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Depth dimension is .29 inch (7.4 mm) for extractors from CAGE 18915 and 5GB68 and .27 inch (6.9 mm) from CAGE 61081.

FIGURE 3. Configuration M dimensions (left hand with bent leg).

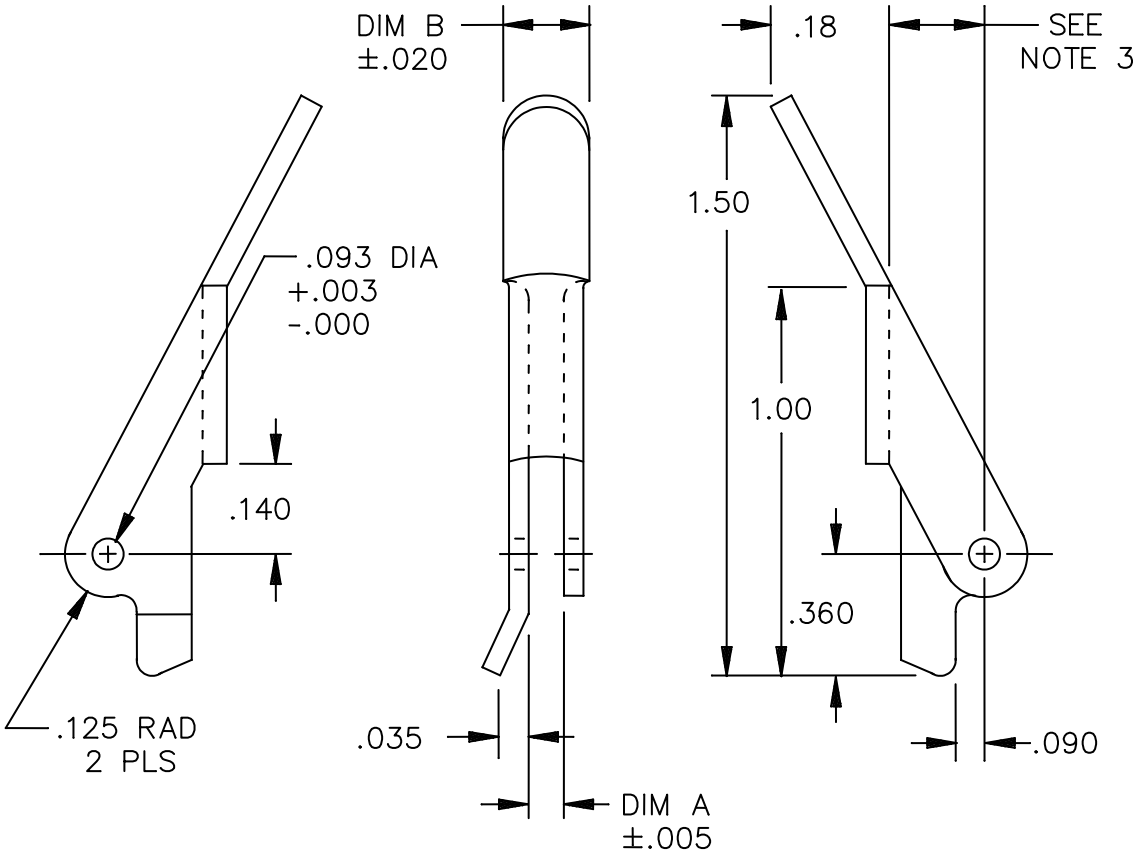
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NOTES:
1. The SED symbol displayed on the extractor handles shall be in accordance with A–A–59832.
2. Unless otherwise specified, a monochromatic reproduction of the symbol displayed in A–A–59832 using any color that contrasts with the color of the extractor handle may be used.

FIGURE 4. Configurations N and P with SED symbol on extractor handle (board edge view).

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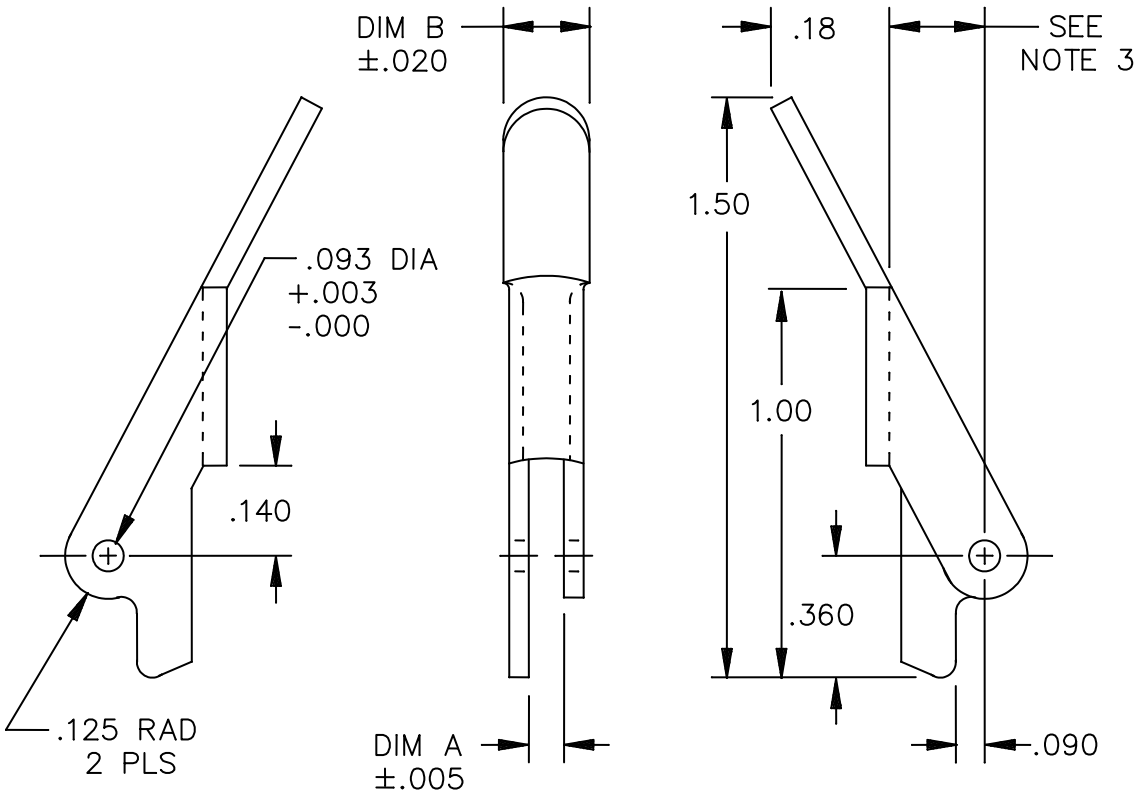
Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.020	0.51	.090	2.29	.125	3.18	.360	9.14
.005	0.13	.035	0.89	.093	2.36	.140	3.56	1.00	25.4
						.18	4.57	1.50	38.1

NOTES:

1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Depth dimension is .29 inch (7.4 mm) for extractors from CAGE 18915 and 5GB68 and .27 inch (6.9 mm) from CAGE 61081.

FIGURE 5. Configuration Q dimensions (right hand with bent leg).

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BOARD EDGE VIEW

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.08	.020	0.51	.093	2.36	.140	3.56	.360	9.14
.005	0.13	.090	2.29	.125	3.18	.18	4.57	1.00	25.4
								1.50	38.1

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. Depth dimension is .29 inch (7.4 mm) for extractors from CAGE 18915 and 5GB68 and .27 inch (6.9 mm) from CAGE 61081.

FIGURE 6. Configuration R dimensions (right hand).

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APPLICATION DATA.

Extractor orientation with circuit card assembly printed board. Figures 7 and 8 show the extractor mounted to a circuit card assembly and installed into a locking card guide from different views. Figure 7 depicts one strait on and two angled views of a right hand extractor fastened to the printed board of a circuit card assembly. Figure 8 depicts a right side view of a circuit card assembly with a right hand extractor fastened to the printed board.

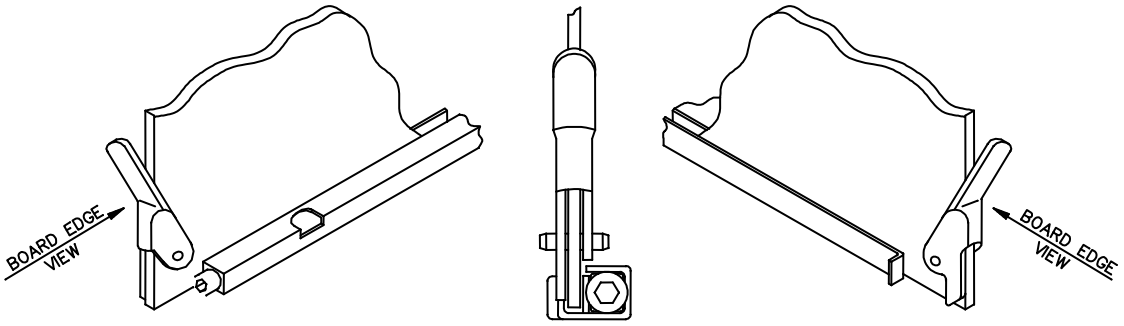


FIGURE 7. Extractor fastened to circuit card assembly (right hand extractor shown).

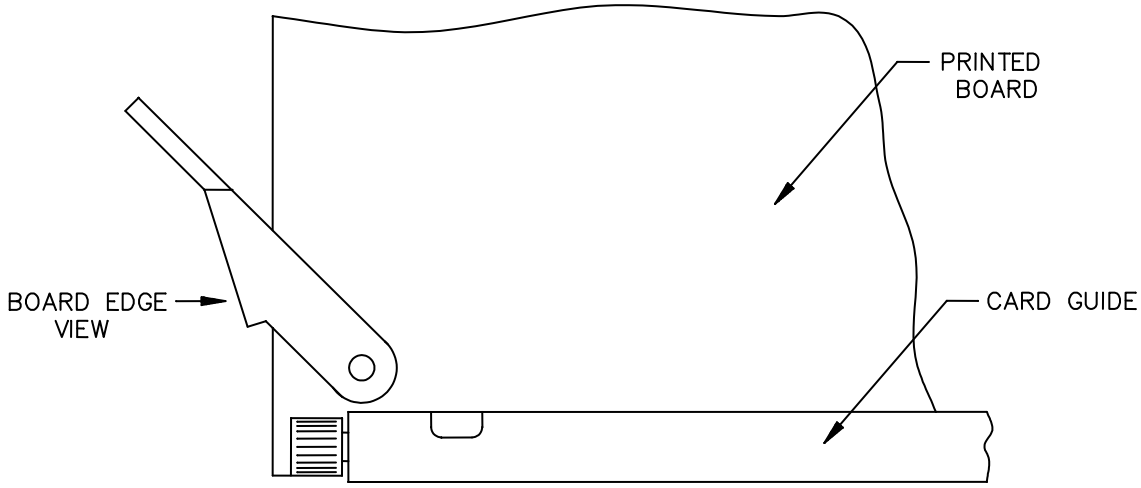
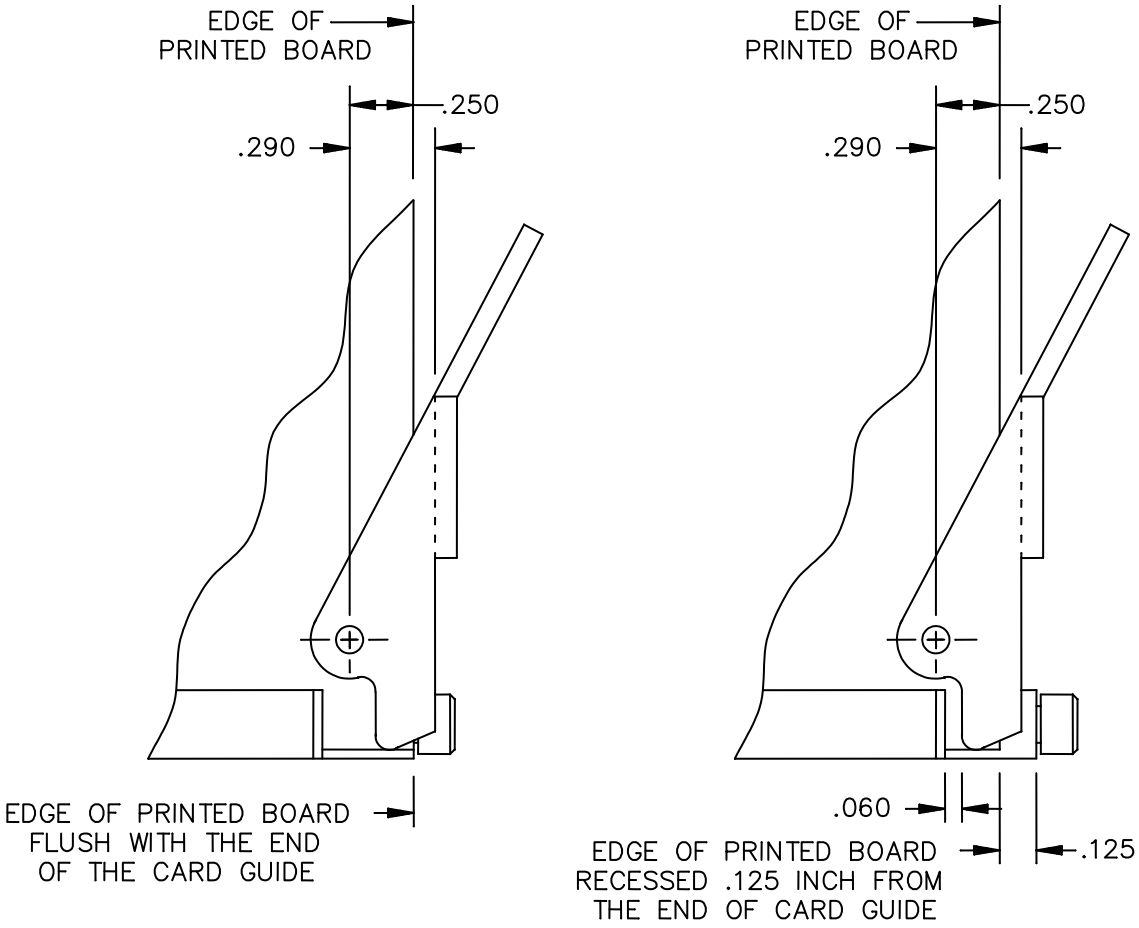


FIGURE 8. Circuit card assembly to card guide view (right hand extractor shown).

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Circuit card positioning in relation to actuating surface. Two positions of a circuit card assembly edge to a locking card guide when seated in the connector are shown on figure 9. The circuit card assembly is ejected from its installed position by lever action of the extractor against the tab on a locking card guide (shown) or actuating surface of a card cage. The extractor will allow .030 inch (0.8 mm) over travel for tolerance take-up. Two extractors mounted on each circuit card assembly is recommended.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.060	1.52	.125	3.18	.250	6.35	.290	7.37

NOTES:
1. Dimensions are in inches. Millimeter equivalents are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.

FIGURE 9. Circuit card assembly to locking card guide dimensions (right hand extractor shown).

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Circuit card assembly printed board. The circuit card assembly should be designed so that electronic components on the printed board in the area around the extractor mounting hole have sufficient clearance for extractor rotation as shown on figure 10.

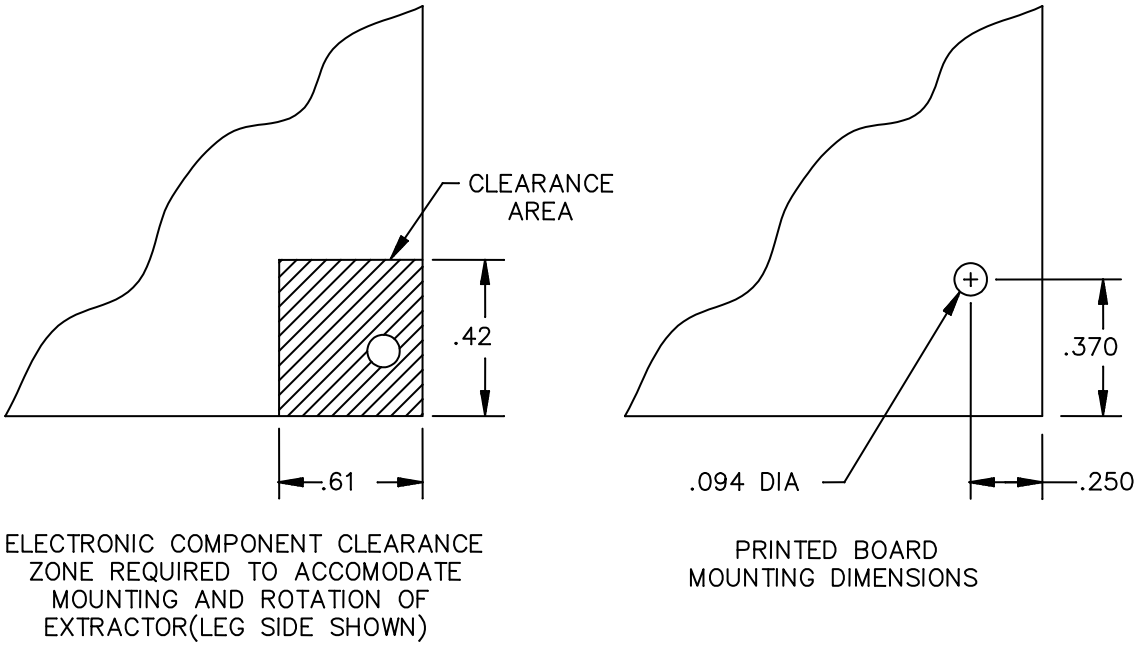


FIGURE 10. Printed board clearance zone and mounting hole dimensions.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59832 – Extractor, Electrical Card, Metal, General Requirements For.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59832.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

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Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher – A Division of Pentair Technical Products 7328 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 (URL): www.birtcherproducts.com
61081	Calmark – A Division of Pentair Technical Products 7328 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 Electronic mail: sales@calmark.com URL: www.calmark.com
5BG68	Card Locks Unlimited, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806–1231	Telephone: (714) 738–6194 Facsimile: (714) 446–0119 E-mail: sales@clumfg.com URL: www.clumfg.com

Part number (P/N) supersession data. This CID specification sheet PINs supersedes the following manufacturer's P/Ns as shown in table IV and on DESC Drawing 84191 (table V) as shown. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE IV. Commercial part number supersession data.

PIN designator AA59832/05	Vendor similar designator or type part number 1/ CAGE 5BG68	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 61081
K11AR	71–T6–1–L–1–E		
K21AR	71–T6–2–L–1–E		1119A–2LRAE
K31AR	71–T6–3–L–1–E		1119A–3LRAE
K41AR	71–T6–4–L–1–E		1119A–4LRAE
K51AR	71–T6–5–L–1–E		1119A–5LRAE
K12AB	71–1–L–5–E		
K22AB	71–2–L–5–E	109–L–BLK–SED	1109A–2LBAE
K32AB	71–3–L–5–E	109–3L–BLK–SED	1109A–3LBAE
K42AB	71–4–L–5–E	109–4L–BLK–SED	1109A–4LBAE
K52AB	71–5–L–5–E		1109A–5LBAE
K11LG	71–T6–1–L–4–E		
K21LG	71–T6–2–L–4–E		1119A–2LCGE
K31LG	71–T6–3–L–4–E		1119A–3LCGE
K41LG	71–T6–4–L–4–E		1119A–4LCGE
K51LG	71–T6–5–L–4–E;		1119A–5LCGE

See note at end of table.

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TABLE IV. Commercial part number supersession data – Continued.

PIN designator AA59832/05	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 61081	Vendor similar designator or type part number 1/ CAGE 5BG68
K12LC	71–1–L–CC		
K22LC	71–2–L–CC	109–L–CLR–SED	1109A–2LCCE
K32LC	71–3–L–CC	109–3L–CLR–SED	1109A–3LCCE
K42LC	71–4–L–CC	109–4L–CLR–SED	1109A–4LCCE
K52LC	71–5–L–CC		1109A–5LCCE
K27PC	71–CR–2–L–4	109CR–L–SED	1129A–2LE
K37PC		109CR–3L–SED	1129A–3LE
K47PC		109CR–4L–SED	1129A–4LE
K57PC			1129A–5LE
L11AR	71–T6–1–L–1		
L21AR	71–T6–2–L–1		1119A–2LRA
L31AR	71–T6–3–L–1		1119A–3LRA
L41AR	71–T6–4–L–1		1119A–4LRA
L51AR	71–T6–5–L–1		1119A–5LRA
L12AB	71–1–L–5		
L22AB	71–2–L–5	109–L–BLK	1109A–2LBA
L32AB	71–3–L–5	109–3L–BLK	1109A–3LBA
L42AB	71–4–L–5	109–4L–BLK	1109A–4LBA
L52AB	71–5–L–5		1109A–5LBA
L11LG	71–T6–1–L–4		
L21LG	71–T6–2–L–4		1119A–2LCG
L31LG	71–T6–3–L–4		1119A–3LCG
L41LG	71–T6–4–L–4		1119A–4LCG
L51LG	71–T6–5–L–4		1119A–5LCG
L12LC	71–1–L–CC		
L22LC	71–2–L–CC	109–L–CLR	1109A–2LCC
L32LC	71–3–L–CC	109–3L–CLR	1109A–3LCC
L42LC	71–4–L–CC	109–4L–CLR	1109A–4LCC
L52LC	71–5–L–CC		1109A–5LCC

See note at end of table.

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TABLE IV. Commercial part number supersession data – Continued.

PIN designator AA59832/05	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 61081	Vendor similar designator or type part number 1/ CAGE 5BG68
L27PC	71–CR–2–L–4	109CR–L	1129A–2L
L37PC		109CR–3L	1129A–3L
L47PC		109CR–4L	1129A–4L
L57PC			1129A–5L
M11AR	71–T6–1–L–1–BL		
M21AR	71–T6–2–L–1–BL		1119A–2LRAD
M31AR	71–T6–3–L–1–BL		1119A–3LRAD
M41AR	71–T6–4–L–1–BL		1119A–4LRAD
M51AR	71–T6–5–L–1–BL		1119A–5LRAD
M12AB	71–1–L–5–BL		
M22AB	71–2–L–5–BL		1109A–2LBAD
M32AB	71–3–L–5–BL		1109A–3LBAD
M42AB	71–4–L–5–BL		1109A–4LBAD
M52AB	71–5–L–5–BL		1109A–5LBAD
M11LG	71–T6–1–L–4–BL		
M21LG	71–T6–2–L–4–BL		1119A–2LCGD
M31LG	71–T6–3–L–4–BL		1119A–3LCGD
M41LG	71–T6–4–L–4–BL		1119A–4LCGD
M51LG	71–T6–5–L–4–BL		1119A–5LCGD
M12LC	71–1–L–CC–BL		
M22LC	71–2–L–CC–BL		1109A–2LCCD
M32LC	71–3–L–CC–BL		1109A–3LCCD
M42LC	71–4–L–CC–BL		1109A–4LCCD
M52LC	71–5–L–CC–BL		1109A–5LCCD
M27PC	71–CR–2–L–4–BL		1129A–2LD
M37PC			1129A–3LD
M47PC			1129A–4LD
M57PC			1129A–5LD

See note at end of table.

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TABLE IV. Commercial part number supersession data – Continued.

PIN designator AA59832/05	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 61081	Vendor similar designator or type part number 1/ CAGE 5BG68
N11AR	71–T6–1–L–1–BL–E		
N21AR	71–T6–2–L–1–BL–E		1119A–2LRADE
N31AR	71–T6–3–L–1–BL–E		1119A–3LRADE
N41AR	71–T6–4–L–1–BL–E		1119A–4LRADE
N51AR	71–T6–5–L–1–BL–E		1119A–5LRADE
N12AB	71–1–L–5–BL–E		
N22AB	71–2–L–5–BL–E		1109A–2LBADE
N32AB	71–3–L–5–BL–E		1109A–3LBADE
N42AB	71–4–L–5–BL–E		1109A–4LBADE
N52AB	71–5–L–5–BL–E		1109A–5LBADE
N11LG	71–T6–1–L–4–BL–E		
N21LG	71–T6–2–L–4–BL–E		1119A–2LCGDE
N31LG	71–T6–3–L–4–BL–E		1119A–3LCGDE
N41LG	71–T6–4–L–4–BL–E		1119A–4LCGDE
N51LG	71–T6–5–L–4–BL–E		1119A–5LCGDE
N12LC	71–1–L–CC–BL–E		
N22LC	71–2–L–CC–BL–E		1109A–2LCCDE
N32LC	71–3–L–CC–BL–E		1109A–3LCCDE
N42LC	71–4–L–CC–BL–E		1109A–4LCCDE
N52LC	71–5–L–CC–BL–E		1109A–5LCCDE
N27PC	71–CR–2–L–4–BL–E		1129A–2LDE
N37PC			1129A–3LDE
N47PC			1129A–4LDE
N57PC			1129A–5LDE

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59832.

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TABLE V. Supersession information.

Superseded (old) PIN in accordance with DESC Drawing 84191	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DESC Drawing 84191	Superseding (new) CID PIN
84191–33ET	AA59832/05 K51LG	84191–34ET	AA59832/05 S51LG
84191–33EA	AA59832/05 K51LG	84191–34EA	AA59832/05 S51LG
84191–33PS	AA59832/05 L57PC	84191–34PS	AA59832/05 R57PC
84191–33CT	AA59832/05 L51LG	84191–34CT	AA59832/05 R51LG
84191–33CA	AA59832/05 L52LG	84191–34CA	AA59832/05 R52LG
84191–33*T 1/	AA59832/05 L51A# 2/	84191–34*T 1/	AA59832/05 R51A# 2/
84191–33*A 1/	AA59832/05 L52A# 2/	84191–34*A 1/	AA59832/05 R52A# 2/
84191–33PSBL	AA59832/05 M57PC	84191–34PSBL	AA59832/05 Q57PC
84191–33CTBL	AA59832/05 M51LG	84191–34CTBL	AA59832/05 Q51LG
84191–33CABL	AA59832/05 M52LG	84191–34CABL	AA59832/05 Q52LG
84191–33*TBL 1/	AA59832/05 M51A# 2/	84191–34*TBL 1/	AA59832/05 Q51A# 2/
84191–33*ABL 1/	AA59832/05 M52A# 2/	84191–34*ABL 1/	AA59832/05 Q52A# 2/
84191–33ETBL	AA59832/05 N51LG	84191–34ETBL	AA59832/05 P51LG
84191–33EABL	AA59832/05 N51LG	84191–34EABL	AA59832/05 P51LG

1/ The asterisk "*" denotes the anodize aluminum finish colors for DESC Drawing 84191.
2/ The pound-sign "#" denotes the anodize aluminum finish colors of table II herein.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

Project: 5998–2010–055

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/.

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